

too easy by the reviewer: “none of the chosen datasets (MATH, GSM8K, and MMLU) uses complicated math concepts”.

3. **Missing or inappropriate baselines.** For example: “The proposed method needs to be compared to simply asking the model to think of one (or several) facts about the question before answering using more turns. This could be an additional baseline to verify the scoring process is meaningful.” and “Although the proposal includes some baselines that should be compared to, it does not mention some methods which seem to do quite well with LLMs.” Sometimes, “the chosen baselines may not be suitable”, for example, because they are not directly comparable with the proposed method.
4. **Making unrealistic assumptions.** For example: “The assumption that model can (mostly) accurately flag its own hallucinations is quite tricky.”, “there is a presupposed assumption that hallucinations in LLMs are ungrounded and independent of the data they are trained on, which is generally not considered true”, “The big issue for the effectiveness of the proposed method is that, it asserts very strong assumptions on downstream tasks, such as there must exist only two extremes.”, “Some assumptions (e.g., [...]) are unlikely to be true in practice, especially when low-resource languages and less represented cultures are included in the study.”, and “A major assumption in this approach is that the model is able to [...]. However, [...]”.
5. **Being too resource-demanding.** Despite the fact that we explicitly prompted the agent to consider feasibility when generating ideas, some of the generated ideas are still too resource-demanding. For example, one reviewer noted: “The biggest issue to feasibility I see is that the project calls for fine-tuning BLOOM (See step 5). BLOOM has 176B parameters so it’s going to take quite a lot of GPUs to fine-tune. From a systems perspective, I see this as causing delays.” In some other cases, manual data annotation is being criticized for feasibility: “The bottleneck seems to be the dataset collection process if there are no existing datasets that fit the requirements of the paper.”, and “the manual evaluation by native speakers or cultural experts could be time-consuming and resource-intensive”.
6. **Not well-motivated.** For example: “Not well-motivated and there is not a clear intuition that this work can work to increase the factuality.”, “And in general the method is not well-motivated and needs reasons why retrieving from model itself is meaningful by use cases or specific tasks.”, and “The idea simply doesn’t make sense to me. Given current LLMs’ ability, I’m pretty sure they can simply recite code like inserting data to a binary search tree.”
7. **Not adequately following existing best practices.** For example: “The proposal does not seem to include awareness of what has been previously tried, or more strategic ways to evaluate success/failures...”

We contrast these with some of the unique strengths and weaknesses of human ideas:

1. **Human ideas are generally more grounded in existing research and practical considerations, but may be less innovative.** For example, these ideas might be applying existing techniques to new problems: “Multilinguality as a debiasing method has already been considered in the literature, although not necessarily in the prompt engineering framework.” Sometimes people apply incremental changes to existing techniques: “The overall idea shares quite a similar idea with program-of-thought (PoT). The only difference is that there is an additional step where an LLM is prompted to decide whether to use code or not.” Some ideas try to combine existing techniques: “Query decomposition and RAG separately are well studied, if there is no existing work that combines both (which I’m not aware of), then it’s reasonably novel.” As some reviewers

noted, human ideas tend to build on known intuitions and results: “There are already existing works on using available lexicons to improve the translation capabilities of LLMs in general.”

2. **Human ideas tend to be more focused on common problems or datasets in the field.** For example: “The problem of models not handling negation properly is a very common problem, especially among propriety LMs such as claude-3-5-sonnet.”, “The data exist. This project mainly entails plugging in these datasets to a prompt template and finetuning for a bit. There is little left unspecified, and it should be quite simple to execute on.”, “I haven’t found any work using this idea to solve this specific problem, but [...] is definitely not new.”, and “While existing works have explored the problem of calibration in long-form answers (e.g. [...]), the specific method for calibration is different.”
3. **Human ideas sometimes prioritize feasibility and effectiveness rather than novelty and excitement.** For example, reviewers noted: “I don’t think this will be a groundbreaking finding, but it will probably work.” and “while the idea is promising and could lead to significant improvements, it may not be groundbreaking enough to be considered transformative or worthy of a best paper award”.

## 8.2 Randomly Sampled Human and AI Ideas with Reviews

We randomly sample four pairs of ideas from different topics to ground our numerical results with actual examples. In each pair, there is one AI idea and one human idea. To save space, we include the full project proposal of each idea along with the full set of reviews in the Appendix, but we list their titles, topics, and average scores here for quick reference (we reveal whether each idea is AI-generated or human-written in Appendix X):

1. Modular Calibration for Long-form Answers: Appendix [P](#)  
Topic: Uncertainty; Average Overall Score: 5.5
2. Semantic Resonance Uncertainty Quantification: Calibrating LLM Confidence through Multi-Path Reasoning: Appendix [Q](#)  
Topic: Uncertainty; Average Overall Score: 6
3. Translation with LLMs through Prompting with Long-Form Context: Appendix [R](#)  
Topic: Multilingual; Average Overall Score: 4
4. Linguistic Pivot Constellation: Enhancing Cross-Lingual Transfer for Low-Resource Languages and Dialects: Appendix [S](#)  
Topic: Multilingual; Average Overall Score: 6.7
5. LLM Directed Retrieval Querying for Improving Factuality: Appendix [T](#)  
Topic: Factuality; Average Overall Score: 4.7
6. Semantic Divergence Minimization: Reducing Hallucinations in Large Language Models through Iterative Concept Grounding: Appendix [U](#)  
Topic: Factuality; Average Overall Score: 3.3
7. Autoprompting: Generate Diverse Few-shot Examples for Any Application: Appendix [V](#)  
Topic: Coding; Average Overall Score: 5
8. Temporal Dependency Unfolding: Improving Code Generation for Complex Stateful Systems: Appendix [W](#)  
Topic: Coding; Average Overall Score: 6.7

## 9 Related Work

**Research idea generation and execution.** Several prior works explored methods to improve idea generation, such as iterative novelty boosting (Wang et al., 2024), multi-agent collaboration (Baek et al., 2024), and multi-module retrieval and revision (Yang et al., 2024). While some of them share similar components as our ideation agent, these works focus on improving the idea generation methods over vanilla prompting baselines, without comparisons to any human expert baselines. Beyond ideation, another line of work uses LLMs for executing experiments by generating code given the research problems (Huang et al., 2024, Tian et al., 2024), or combining idea generation with code generation to directly implement AI-generated ideas (Li et al., 2024, Lu et al., 2024). These works either use automatic evaluation on a pre-defined set of problems and benchmarks, setting a constrained problem space; or rely on proxy metrics like LLM evaluators, which are often unreliable.

**LLM for other research-related tasks.** LLMs have also been used for several other research-related tasks, such as generating code to perform data-driven discovery (Gu et al., 2024, Guo et al., 2024, Hu et al., 2024, Ifargan et al., 2024, Majumder et al., 2024), automatic review generation (D’Arcy et al., 2024, Liang et al., 2024), related work curation (Ajith et al., 2024, Kang and Xiong, 2024, Lehr et al., 2024, Press et al., 2024), experiment outcome prediction (Hewitt et al., 2024, Lehr et al., 2024, Manning et al., 2024, Zhang et al., 2024), and future work recommendation (Zhang et al., 2024). Unlike these works, we tackle the more creative and open-ended task of research ideation.

**Computational creativity.** Our work also connects to the line of work on examining AI’s novelty and diversity in creative tasks. Chakrabarty et al. (2024) found that AI writings are less creative than professional writers, while we show LLM-generated ideas can be more novel than experts on the task of research ideation. Another line of work found that LLM generations lack collective diversity (Anderson et al., 2024, Zhou et al., 2024), which matches our findings on idea generation. Lastly, several other works conducted human evaluation to study the impact of AI exposure or human-AI collaboration on novelty and diversity (Ashkinaze et al., 2024, Liu et al., 2024, Padmakumar and He, 2024) with mixed conclusions. While we also conduct a human evaluation of idea novelty, we focus on the human-AI comparison on the challenging task of research ideation with expert participants.

## 10 Discussion

To summarize, we compared research ideas generated by our AI agent with ideas written by expert researchers, and observed the robust finding that expert reviewers rate AI ideas as statistically more novel than expert ideas. In this section, we discuss some high-level questions readers might have and suggest some ways to address them.

**Question 1: Do these collected expert ideas represent their best ideas?** One might argue that these ideas submitted by our idea-writing participants might not represent their best ideas as we discussed in Section 6.1, since most of them came up with the idea on the spot within a short period. In order to address this concern, we have designed an experiment where we will compare AI ideas with papers accepted at top-tier AI conferences. To avoid any possible contamination, we target the upcoming EMNLP 2024 conference, which will release the accepted papers in October 2024. We have generated AI ideas with our agent on 23 topics from the EMNLP Call For Papers page in July 2024 and cached them. We pre-registered our analysis plan which also includes the link to the cached ideas.<sup>12</sup> Apart from comparing the quality of these ideas, we will also compute the overlap between AI-generated ideas and accepted papers on the same topics.

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<sup>12</sup><https://osf.io/z6qa4>

**Question 2: Are evaluations based solely on ideas subjective?** In this current study, we focused solely on evaluating the ideas themselves. Ideas that sound novel and exciting might not necessarily turn into successful projects, and our results indeed indicated some feasibility trade-offs of AI ideas. We view the current study as a preliminary evaluation of AI-generated ideas. In the next phase, we will recruit researchers to execute some AI and human-generated ideas into full projects. This will enable reviewers to assess the complete experimental outcomes, providing a more reliable basis for evaluation. Furthermore, it will allow us to analyze whether our initial idea evaluations align with the assessments of the actual project outcomes.

**Question 3: Why do you focus only on prompting-based research in NLP?** The scope of our study is limited to prompting research ideas within NLP. We chose this design to facilitate the next phase of our execution experiment, where we prefer research ideas that are less resource-demanding and can be executed relatively quickly. We believe that the evaluation protocols we established should be applicable to other research domains as well, although the conclusions could be different depending on the research fields. Future work should consider extending such human study to other research domains and it would be interesting to compare how the conclusions differ.

**Question 4: Can you automate idea execution as well?** It is tempting to envision an end-to-end automated research pipeline where AI agents can implement AI-generated ideas to directly evaluate their effectiveness. Apart from speeding up scientific discovery, one could also imagine using such execution agents to automatically verify experiment results in existing papers or new submissions. We have also explored building an LLM agent to generate code to implement the generated ideas. Specifically, we provide a template codebase that consists of: (1) loading datasets from Huggingface or generating synthetic test examples; (2) implementing baseline methods; (3) implementing the proposed method; (3) loading or implementing the evaluation metrics; (4) running experiments on the testset with the baselines and the proposed method, so that the output of the agent will be a report of the baseline performance as well as the proposed method’s performance. While this agent can generate code that compiles and executes, we find that the automated experiments can be **misleading** because the agent often skips or modifies steps in the baselines or proposed methods. In some cases, the metric functions are also not correctly defined. This highlights the core challenge: just comparing the final experiment results is not enough; we have to verify the faithfulness of the implementations as well. Performing such implementation verification is not a trivial task, and we leave it to future work. We provide detailed description of our idea execution agent in Appendix Y.

## 11 Ethical Considerations

**Publication Policy.** The growing use of AI to generate research ideas raises serious concerns about the potential abuse of these technologies by students or researchers who may flood academic conferences with low-quality or poorly thought-out submissions. The availability of LLM-generated content could lead to a decline in the overall quality of academic discourse, as some individuals might take a lazy approach, relying on AI to both generate ideas and review submissions. This would undermine the credibility and integrity of the review process. The risks are real. Without proper oversight, we could see a deluge of submissions that lack depth or intellectual merit. To prevent this, it is essential to hold researchers accountable for the outputs generated through AI tools. Rigorous standards must be applied equally to both AI-assisted and human-generated research to ensure that the use of LLMs does not result in misleading, superficial, or unethical academic contributions.

**Intellectual Credit.** The use of LLMs to generate research ideas introduces significant ambiguity around the concept of intellectual credit. Traditional frameworks for attributing credit in research, based on human authorship and contribution, become less clear when AI plays a significant role

in idea generation. Questions arise around how to distribute credit between the developers of the LLM, the researchers who designed the frameworks for its use, and the researchers who integrate AI-generated ideas into their work. Furthermore, it becomes increasingly difficult to trace the origins of AI-generated contributions, especially when they draw from vast datasets composed of numerous sources. This complexity calls for a broader rethinking of how intellectual credit is assigned in AI-driven research. While a complete overhaul of legal and academic norms is beyond the scope of this project, we advocate for the adoption of transparent documentation practices. Researchers should clearly disclose the role AI played in the idea generation process, specifying which models, data sources, and frameworks were used, and outlining the level of human involvement. This could ensure that the credit distribution in AI-supported research is as transparent and fair as possible.

**Potential for Misuse.** AI-generated research ideas, especially those that introduce novel concepts, have the potential to be misused in ways that could lead to harmful or destabilizing outcomes. For instance, ideation agents could be leveraged to generate adversarial attack strategies or other unethical applications. This concern aligns with broader arguments from those focused on existential risk (X-risk), who argue that AI-driven innovation could be a primary route to destabilizing the status quo, posing risks at a societal or even global level. Our stance is that such discussions on safety should be evidence-based to the extent that it is possible, and careful evaluation work is an important component of keeping these discussions grounded in actual, measured capabilities of these systems. We advocate for continued safety research specifically targeting these types of concerns—such as the development of Reinforcement Learning from Human Feedback (RLHF) systems or anti-jailbreak mechanisms for research ideation agents. Additionally, we believe it would be meaningful to create safety benchmarks that assess the ethical and safe application of AI-generated ideas.

**Idea Homogenization.** Our analysis showed that current LLMs lack diversity in idea generation. This raises important concerns that wide adoption of LLMs can result in idea homogenization, where the generated ideas only reflect a narrow set of perspectives or have systematic biases. Over time, this could lead to a reduction in the richness and diversity of research outputs globally. Future work should develop ways to either improve LLMs themselves or refine our idea generation methods to promote idea diversity. It’s also important to note that our evaluation primarily assesses the quality of the typical ideas being generated, and may not fully capture the long tail of unique or novel ideas that would be truly transformative.

**Impact on Human Researchers.** The integration of AI into research idea generation introduces a complex sociotechnical challenge, as research is fundamentally a community-driven, collaborative effort. By introducing AI, particularly LLMs, into this social system, we risk unforeseen consequences. Overreliance on AI could lead to a decline in original human thought, while the increasing use of LLMs for ideation might reduce opportunities for human collaboration, which is essential for refining and expanding ideas. To mitigate these risks, future works should explore new forms of human-AI collaboration, and our results on human reranking of AI ideas show that even naive human-AI collaboration approaches can be effective. Beyond reranking, humans can play a critical role in the ideation process by providing intermediate feedback, taking AI-generated ideas as inspiration for further development, and bringing their unique expertise into the process. Understanding how to integrate LLMs into this collaborative process without disrupting the social fabric of research will be an important ongoing problem, requiring careful consideration of the broader sociotechnical implications.